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LAB TASK#2

QUESTION#1:

```
#include<iostream>
#include<cstdlib>
#include<ctime>

using namespace std;

float Calculate_average_stock(int size, int* inventory) {
int total_stock = 0;

for (int i=0;i<size;i++) {
total_stock += inventory[i];
}
return static_cast<float>(total_stock)/size; // type casting to convert int to float for precision

}

int Critical_products(int size,int* inventory){

int avg= Calculate_average_stock(size,inventory);
int count = 0;
for (int i=0;i<size;i++) {
if (inventory[i]<avg) {
++count;
}
}
return count;

}


int Find_Top_Selling_Products(int size,int* inventory){

int maxIndex = 0;
for (int i=1;i<size;i++) {
if (inventory[i] > inventory[maxIndex]) {
maxIndex = i;
}
}
return maxIndex;

}

int Find_Second_Best_Selling_Products(int size,int *inventory){
int maxIndex = 0;
int secondMaxIndex = -1;
for (int i=1;i<size;i++) {
if (inventory[i]>inventory[maxIndex]) {
secondMaxIndex=maxIndex;
maxIndex=i;
```

```
} else if (secondMaxIndex==-1 || inventory[i]<inventory[secondMaxIndex]) {
secondMaxIndex = i;
}
}
return secondMaxIndex;

}


void Sorting_By_popularity(int size,int *inventory){
for (int i=0;i<size-1;i++) {
for (int j=0;j<size-i-1;j++) {
if (inventory[j] < inventory[j+1]) {
// Swap if the current element is less than the next element
return swap(inventory[j], inventory[j + 1]);
}
}
}
}


int main(){

int size,choice;

cout<<"enter the size if the inventory array"<<endl;
cin>>size;

srand(time(NULL)); //random generation with time

int *inventory=new int[size];

for(int i=0;i<size;i++){
inventory[i]= rand()%100+1; //storing the random numbers in array
}

cout<<"1. Calculate the average stock level"<<endl<<"2. Identify critical products"<<endl<<"3.
Discover top-selling product"<<endl<<"4. Find second best-selling product"<<endl<<"5. Sort
products by popularity"<<endl;
cin>>choice;
switch(choice){
case 1:
//calling calulate average function through pointer function
{
float (*fun_average)(int,int*);
fun_average=&Calculate_average_stock;
float res=fun_average(size,inventory);

cout<<"Average Stock level: "<<res<<endl;
break;

}
```

```
case 2:

//calling critical products function through pointer function
{
int (*fun_critical)(int,int*);
fun_critical=&Critical_products;
int res2=fun_critical(size,inventory);
cout<<"Critical products: "<<res2<<endl;
break;
}

case 3:

{ //calling top selling products function through pointer function
int (*fun_top_selling)(int,int*);
fun_top_selling=&Find_Top_Selling_Products;
int res3=fun_top_selling(size,inventory);
cout<<"Top selling product: "<<res3<<endl;
break;
}

case 4:

{ //calling second best selling products function through pointer function
int (*fun_Sec_top_selling)(int,int*);
fun_Sec_top_selling=&Find_Second_Best_Selling_Products;
int res4=fun_Sec_top_selling(size,inventory);
cout<<"Second best selling product: "<<res4<<endl;
break;
}

case 5:

{
//calling sorting function through pointer function
void (*fun_Sort)(int,int*);
fun_Sort=&Sorting_By_popularity;
fun_Sort(size,inventory);
cout<<"Sorted Inventory by Popularity: "<<endl;
for (int i=0;i<size;++i)
{
cout<<inventory[i]<< " ";
}
cout<<endl;

break;
}

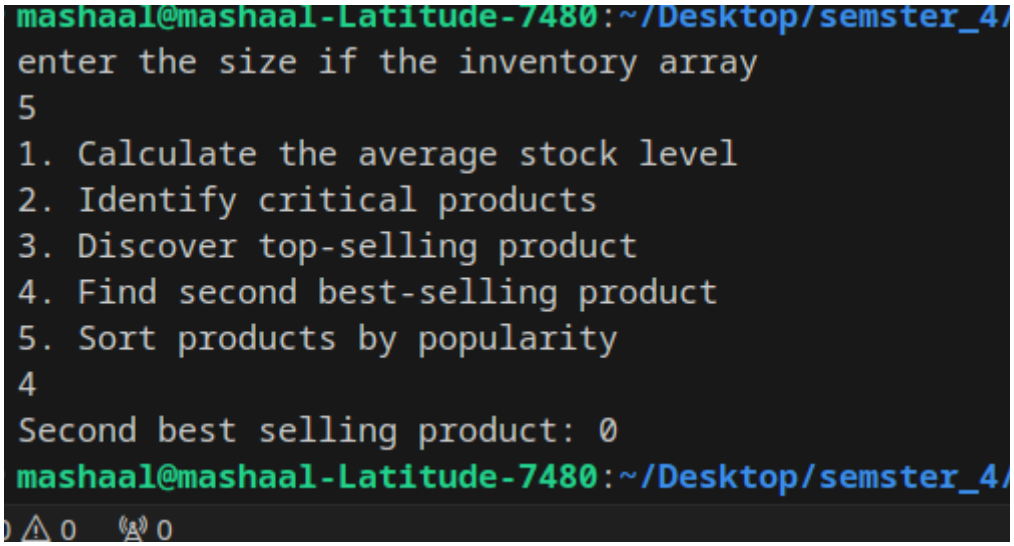

delete[] inventory;
}
```

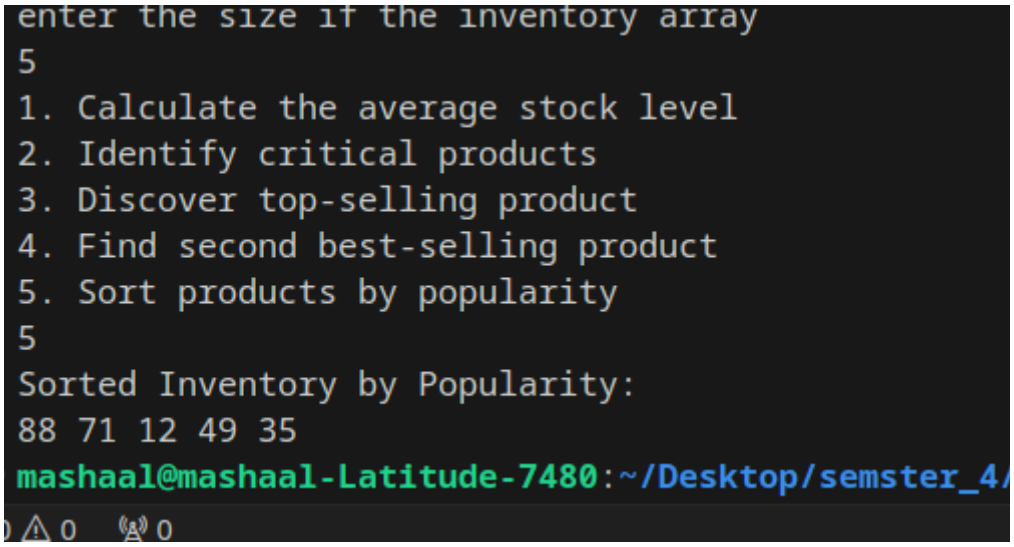
}

```
mashaal@mashaal-Latitude-7480:~/Desktop/semster_4/
enter the size if the inventory array
5
1. Calculate the average stock level
2. Identify critical products
3. Discover top-selling product
4. Find second best-selling product
5. Sort products by popularity
1
Average Stock level: 47.4
mashaal@mashaal-Latitude-7480:~/Desktop/semster_4/
```

```
mashaal@mashaal-Latitude-7480:~/Desktop/semster_4/
enter the size if the inventory array
5
1. Calculate the average stock level
2. Identify critical products
3. Discover top-selling product
4. Find second best-selling product
5. Sort products by popularity
2
Critical products: 4
mashaal@mashaal-Latitude-7480:~/Desktop/semster_4/
```

```
mashaal@mashaal-Latitude-7480:~/Desktop/semster_4/
enter the size if the inventory array
5
1. Calculate the average stock level
2. Identify critical products
3. Discover top-selling product
4. Find second best-selling product
5. Sort products by popularity
3
Top selling product: 2
mashaal@mashaal-Latitude-7480:~/Desktop/semster_4/
```





QUESTION#2:

```
  #include<iostream>
#include<cstring>

using namespace std;

char* String_between_indexes(char *string,int start_index,int stop_index)
```

```
{
int length = strlen(string);

if(start_index<0 || stop_index>=length || start_index>stop_index){
cout<<"Invalid indices "<<endl;

return 0;
}

int substringLength = stop_index - start_index + 1;

char* result = new char[substringLength + 1];

strncpy(result,string+start_index,substringLength);
return result;

}

int main(){
int max_length=100;
char string[max_length];

cout<<"enter a string"<<endl;
cin.getline(string,max_length);

int start_index,stop_index;

cout<<"Enter start index: ";
cin>>start_index;

cout<<"Enter stop index: ";
cin>>stop_index;

char *result=String_between_indexes(string,start_index,stop_index);

if (result != nullptr) {
cout<<"Result: "<<result<<endl;

delete[] result;
}
}
```

